

National Institute of Textile Engineering and Research (NITER)
B.Sc. in Computer Science & Engineering Level-1, Term-2, 2nd In-Course-2024

Course Title: Developing English Language Skills

Course Code: ENG-1205

Duration: 1 Hour

Answer the following questions:

1. Suppose you are the campus reporter of the newspaper *The Daily Sun*. Write a report to be published in the weekly magazine of this newspaper on the recent program titled "Computer Science Innovations by Rising Scientists" organized by your institution. 15
2. Write a paragraph on "Effects of social media on students' mental health". 10

National Institute of Textile Engineering and Research (NITER)

1st Year 2nd Semester B.Sc. (2nd Class Test) Examination-2024

Department of Computer Science and Engineering (CSE)

MATH 1204: Methods of Integration, Differential Equations and Series

Duration: 1 Hour

Full Marks: 25

Answer all of the following questions.

1. Find $y(2.2)$ using Euler's method from the equation $\frac{dy}{dx} = -xy^2$; $y(2) = 1$ with $h = 0.05$. 10
2. The population of a community is known to increase at a rate proportional to the number of people present at time t . If the population doubles in 50 years, how long will it take to triple? 5
3. Solve any two of the followings, 2*5=10
 - a) $\frac{dy}{dx} = \frac{y}{x} + \tan \frac{y}{x}$ using Homogeneous differential equation.
 - b) $(x^2y - 2xy^2)dx - (x^3 - 3x^2y)dy = 0$ using Exact differential equation.
 - c) $(1+xy)ydx + (1-xy)x dy = 0$ using Exact differential equation.

type-3

1st Year 2nd Semester B.Sc. (2nd Class Test) Examination-2024
Dept. of Computer Science and Engineering
National Institute of Textile Engineering and Research (NITER)

Course Code: CSE-1202

Course Name: Digital Logic Design (3 Credit)

Full Marks: 25

Time: 1 Hour

1. Simplify the following Boolean functions, using K maps: 16

i) $F(x, y, z) = (0, 2, 4, 5, 6)$

ii) $F(w, x, y, z) = (2, 3, 12, 13, 14, 15)$

iii) $F(A, B, C, D) = (0, 6, 8, 13, 14)$ $d(A, B, C, D) = (2, 4, 10)$

iv) $F(x, y, z) = x'yz + xy'z' + xy'z$

2. Find all the prime implicants for the following Boolean functions, and 5

determine which are essential: $F(w, x, y, z) = (0, 2, 4, 5, 6, 7, 8, 10, 13, 15)$

3. Define prime implicants and essential prime implicants. 4

National Institute of Textile Engineering and Research (NITER)

Nayarhat, Savar, Dhaka

B. Sc. in Computer Science and Engineering

1st Year, 2nd Semester, Class Test-2

Course Code: PHY-1203

Course Title: Physics

Full Marks: 25

Time: 60 min

Answer **All** from (1 -6) any **Two** from (7 – 9)

- | | | |
|---|---|---|
| 1 | What is crystalline and non-crystalline solids? | 2 |
| 2 | Define lattice, lattice parameters and lattice constant. | 3 |
| 3 | What is Bravais and non-Bravais lattice | 2 |
| 4 | What is unit cell? Distinguish between primitive and non-primitive unit cell. | 3 |
| 5 | What is co-ordination number? What are the co-ordination number of simple cubic, FCC and BCC lattice? | 2 |
| 6 | What is atomic packing factor (APF)? Calculate APF for simple cubic lattice. | 3 |
| 7 | What is Miller indices? How to find Miller indices? | 5 |
| 8 | For cubic system show that $d_{hkl} = \frac{1}{\sqrt{h^2+k^2+l^2}}$, where the symbols have their usual meaning. | 5 |
| 9 | What is X-ray diffraction? Deduce Bragg's law. | 5 |

National Institute of Textile Engineering and Research

Nayarhat, Savar, Dhaka

B.Sc. in CSE 1st Year 2nd Semester Class Test – 02

Course name: **Fundamentals of Programming**
Marks: 25

Course Code: **CSE - 1201**

Time: 1 Hour

Answer the Following Questions (Any 5)

1. Explain about how one-dimensional array is declared in C and how its elements are laid out in memory. What happens if you access an index outside the valid range of the array? 5
2. Write a C program that declares a string and counting the number of vowels in a string- "***I Love My Country***" and print the counting number as output. 5
3. Discuss about the two-dimensional array in C. Assume an integer type matrix matrix [4][5] and explain how elements are indexed in memory (row-major order). 5
4. Describe about a pointer variable in C. Discuss about how will you declare a pointer, assign it a valid address, and initialize it to a null pointer. 5
5. Compare and contrast using recursion versus an iterative function (loop construct) to solve a problem. 5
6. Write C program to compute the factorial of a non-negative integer using recursion. 5